Practical 4

Class A: Class A IP addresses are used for very large networks and have the first octet as the network part and the remaining three octets as the host part. The prefix of this class is 0.

Class B: Class B IP addresses are used for medium-sized networks and have the first two octets as the network part and the remaining two octets as the host part. The prefix of this class is 10.

Class C: Class C IP addresses are used for small networks and have the first three octets as the network part and the last octet as the host part. The prefix of this class is 110.

→ The IP address `127.0.0.1` is called the *localhost* address. It refers to the same computer on which the network request is being made.

What Is DHCP?

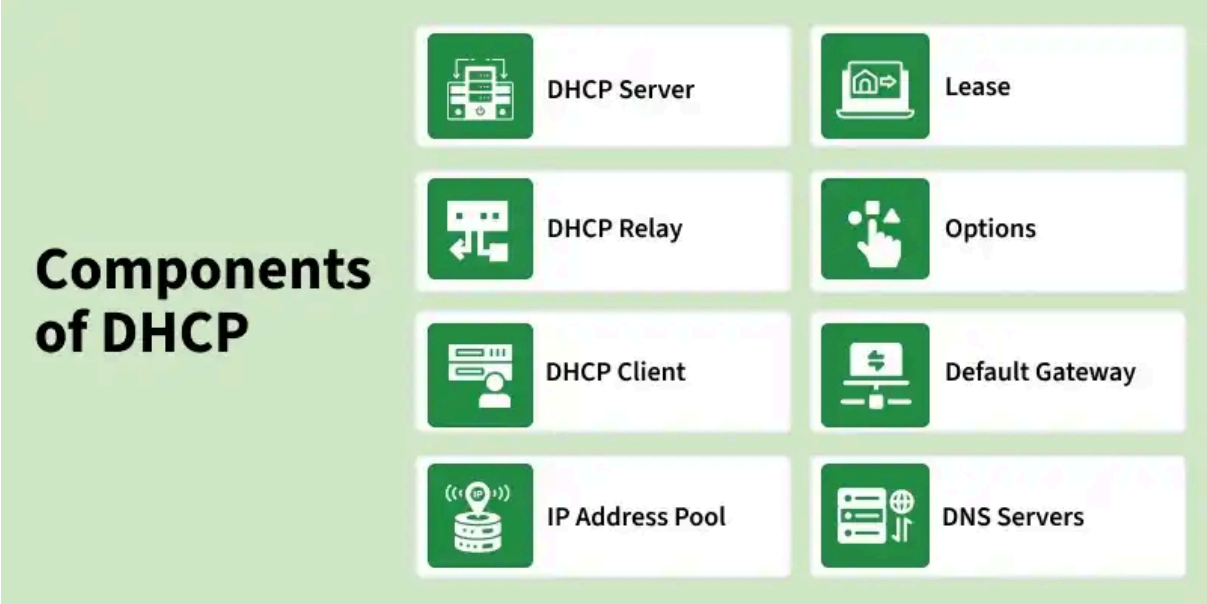
→ DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and other network settings to devices connected to a network.

For example, when your laptop connects to Wi-Fi, DHCP automatically gives it an IP address, subnet mask, default gateway, and DNS server.

Example

Suppose a laptop connects to a Wi-Fi network:

- The laptop sends a DHCP Discover message.
- The router (acting as a DHCP server) assigns:
  - IP address: 192.168.1.20
  - Subnet mask: 255.255.255.0
  - Gateway: 192.168.1.1
  - DNS: 8.8.8.8
- The laptop can now communicate on the network and access the internet.
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What is ARP?

→ ARP (Address Resolution Protocol) is a network protocol used to find the MAC address of a device when its IP address is known on a local network.

In simple words:
→ ARP converts an IP address into its corresponding MAC address.

ARP Command:

The **ARP** command is used to view and manage the ARP table on your computer.

For example, in Windows:

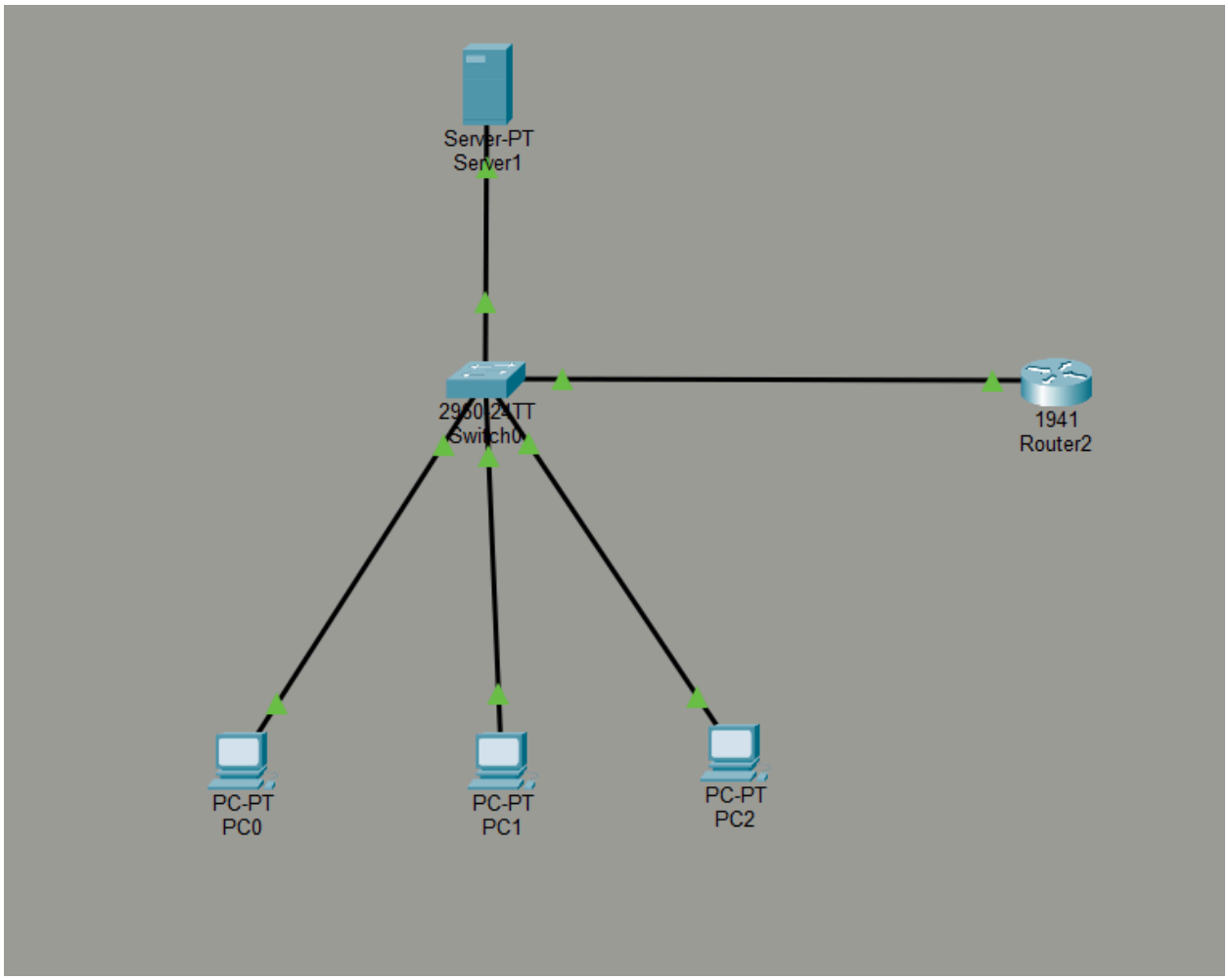
arp -a

→ This displays the ARP table, which contains mappings between IP addresses and MAC addresses.

Problem Definition:

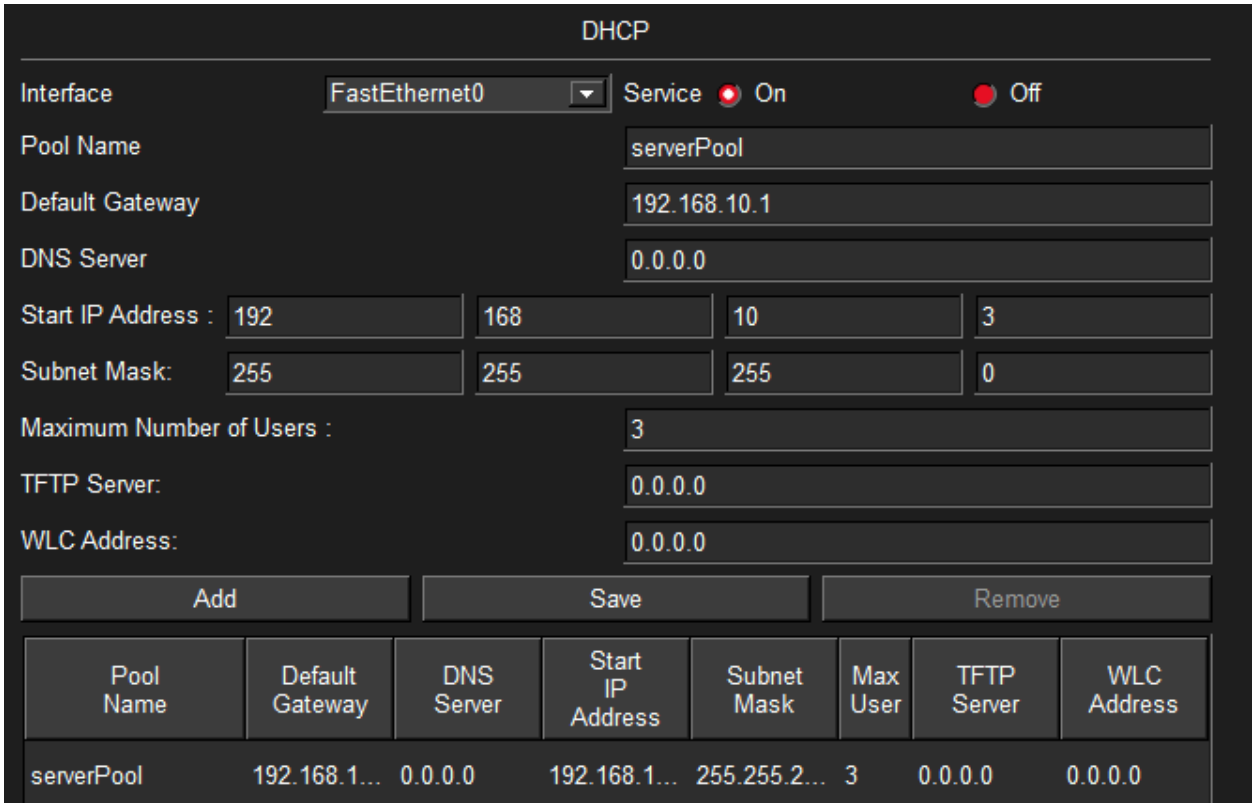
→ Develop experiential understanding of Layer-3 communication by configuring interconnected networks, analyzing router forwarding decisions, routing tables, default gateways, ARP operation and TTL changes, and implementing static routing using Cisco Packet Tracer and Linux-based packet observation.

Solution:



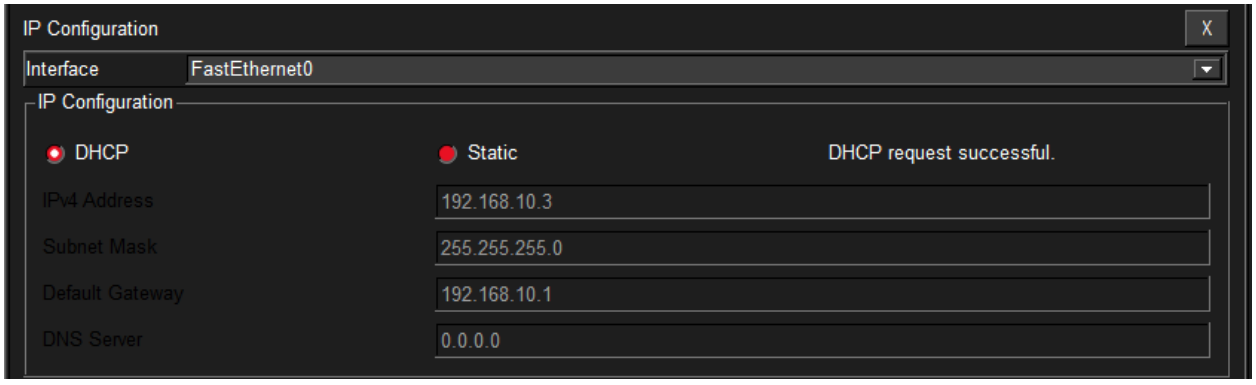
Output:

DHCP Server:

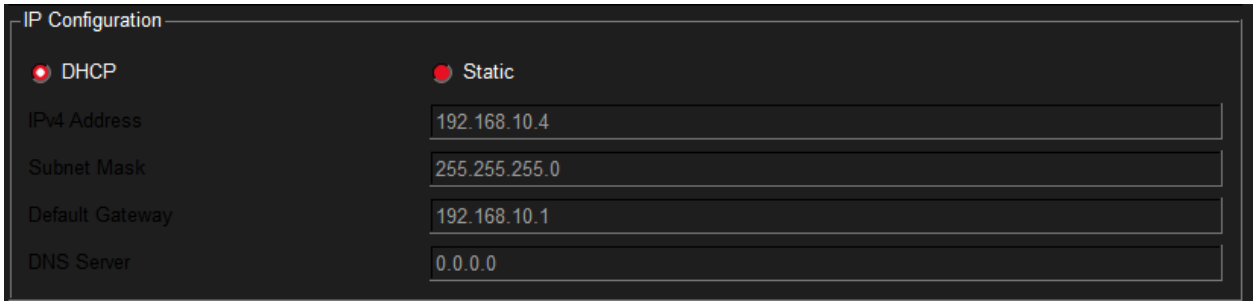


PC0:

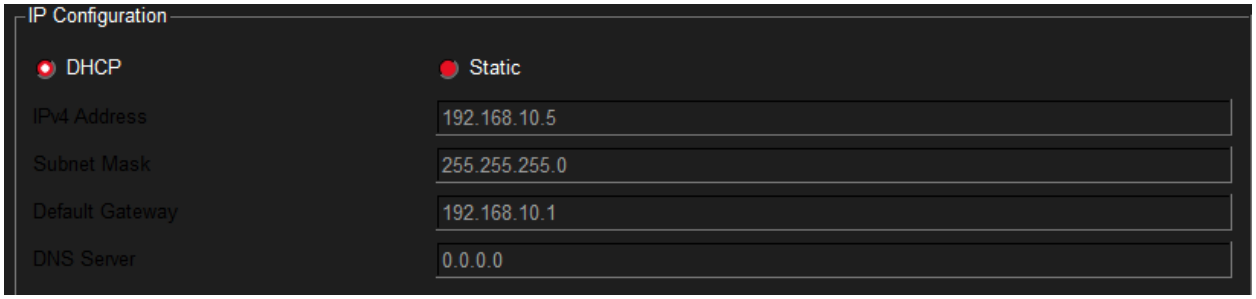
→ IPv4 is given by the DHCP server.



PC1:



PC2:



From PC0 to PC1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

From PC0 TO PC2:

```
C:\>ping 192.168.10.5

Pinging 192.168.10.5 with 32 bytes of data:

Reply from 192.168.10.5: bytes=32 time=1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time=1ms TTL=128
Reply from 192.168.10.5: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Learning:

→ I learned how Layer-3 communication works by configuring interconnected networks and understanding routing tables, default gateways, ARP, and TTL. I also learned how routers forward packets using static routing and observed packet behavior using Cisco Packet Tracer and Linux.